CONCEPT 39.5

Plants respond to attacks by pathogens and herbivores *(pp. 866–869)*

- The **hypersensitive response** seals off an infection and destroys both pathogen and host cells in the region. **Systemic acquired resistance** is a generalized defense response in organs distant from the infection site.
- In addition to physical defenses such as thorns and trichomes, plants produce distasteful or toxic chemicals, as well as attractants that recruit animals that destroy herbivores.

? *How can insects make plants more susceptible to pathogens?*

TEST YOUR UNDERSTANDING

For more multiple-choice questions, go to the **Practice Test** in the **Mastering Biology** eText or Study Area, or go to goo.gl/GruWRg.

Levels 1-2: Remembering/Understanding

1. The hormone that helps plants respond to drought is
(A) auxin.
(B) abscisic acid.
(C) cytokinin.
(D) ethylene.

2. A barley mutant lacking a gibberellic acid receptor would
(A) fail to make GA.
(B) catalyze starch more quickly.
(C) fail to make α-amylase.
(D) fail to take up water.

3. Charles and Francis Darwin discovered that
(A) auxin is responsible for phototropic curvature.
(B) red light is most effective in shoot phototropism.
(C) light destroys auxin.
(D) light is perceived by the tips of coleoptiles.

4. How may a plant respond to *severe* heat stress?
(A) by reorienting leaves to increase evaporative cooling
(B) by creating air tubes for ventilation
(C) by producing heat-shock proteins, which may protect the plant's proteins from denaturing
(D) by increasing the proportion of unsaturated fatty acids in cell membranes, reducing their fluidity

Levels 3-4: Applying/Analyzing

5. The signaling molecule for flowering might be released earlier than usual in a long-day plant exposed to flashes of
(A) far-red light during the night.
(B) red light during the night.
(C) red light followed by far-red light during the night.
(D) far-red light during the day.

6. If a long-day plant has a critical night length of 9 hours, which 24-hour cycle would prevent flowering?
(A) 16 hours light/8 hours dark
(B) 14 hours light/10 hours dark
(C) 4 hours light/8 hours dark/4 hours light/8 hours dark
(D) 8 hours light/8 hours dark/light flash/8 hours dark

7. A plant mutant that shows normal gravitropic bending but does not store starch in its plastids would require a reevaluation of the role of ________ in gravitropism.
(A) auxin
(B) calcium
(C) statoliths
(D) differential growth

8. **DRAW IT** Indicate the response to each condition by drawing a straight seedling or one with the triple response.

	Control	Ethylene added	Ethylene synthesis inhibitor
Wild-type			
Ethylene insensitive (*ein*)			
Ethylene overproducing (*eto*)			
Constitutive triple response (*ctr*)			

Levels 5-6: Evaluating/Creating

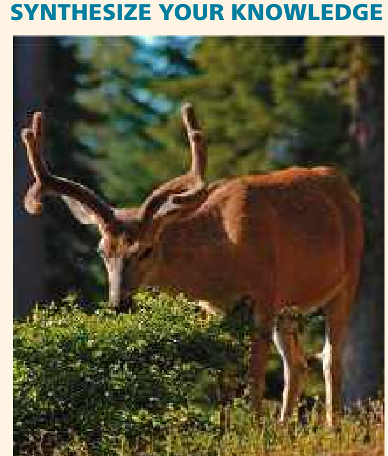
9. **EVOLUTION CONNECTION** In general, light-sensitive germination is more pronounced in small seeds compared with germination of large seeds. Suggest a reason why.

10. **SCIENTIFIC INQUIRY** A plant biologist observed a peculiar pattern when a tropical shrub was attacked by caterpillars. After a caterpillar ate a leaf, it would skip over nearby leaves and attack a leaf some distance away. Simply removing a leaf did not deter caterpillars from eating nearby leaves. The biologist suspected that an insect-damaged leaf sent out a chemical that signaled nearby leaves. How could the researcher test this hypothesis?

11. **SCIENCE, TECHNOLOGY, AND SOCIETY** Describe how our knowledge about the control systems of plants is being applied to agriculture or horticulture.

12. **WRITE ABOUT A THEME: INTERACTIONS** In a short essay (100–150 words), summarize phytochrome's role in altering shoot growth for the enhancement of light capture.

13. **SYNTHESIZE YOUR KNOWLEDGE**



This mule deer is grazing on the shoot tips of a shrub. Describe how this event will alter the physiology, biochemistry, structure, and health of the plant and identify which hormones and other chemicals are involved in making these changes.

For selected answers, see Appendix A.